Executive Dashboard

Operations overview, KPIs and AI executive summary

Test Report

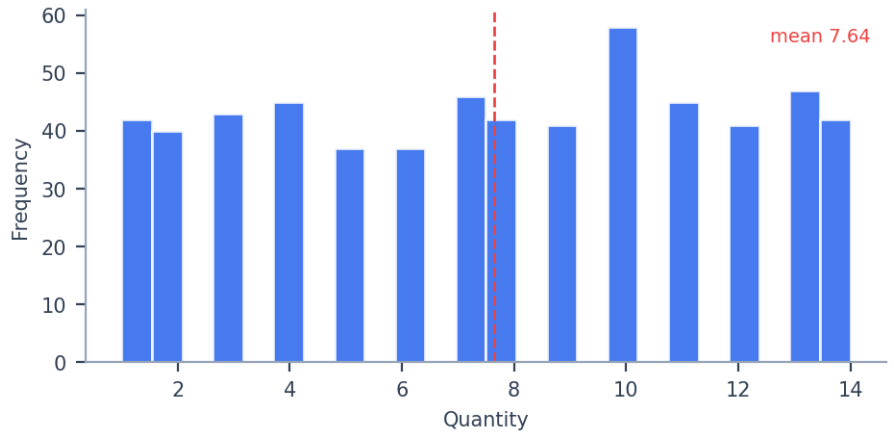
15 Oct 2026, 12:31

OBSERVATIONS	FEATURES	COMPLETENESS	DUPLICATES	BEST MODEL R2	SIGNIFICANT TESTS
606	11	99.43%	6	0.931	4
rows analysed	8 numeric / 3 categorical	38 missing cells	duplicate rows	Linear Regression	ANOVA + chi-square (p<0.05)

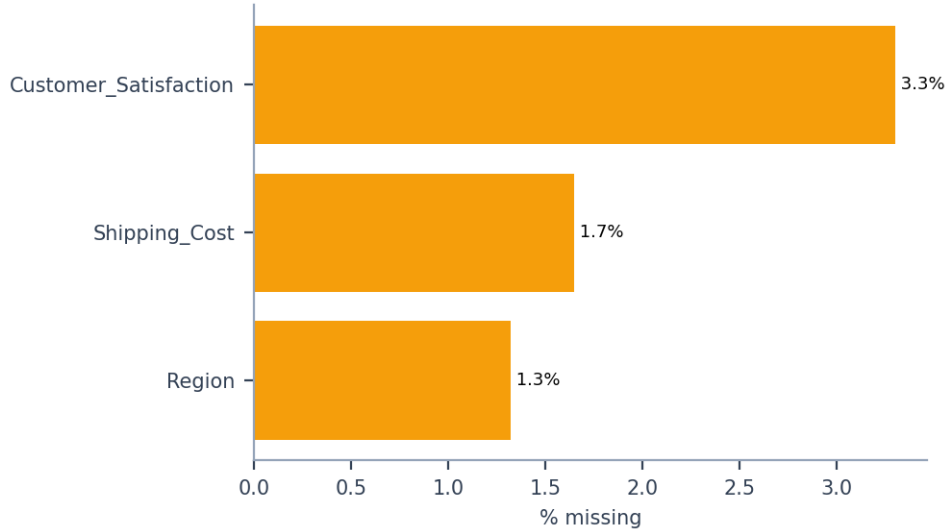
AI Executive Summary

This report analyses 606 records across 11 fields. Data completeness stands at 99.43% and 6 duplicate rows were detected. A Linear Regression model explains 93.1% of the variance in 'Sales', indicating strong predictability.

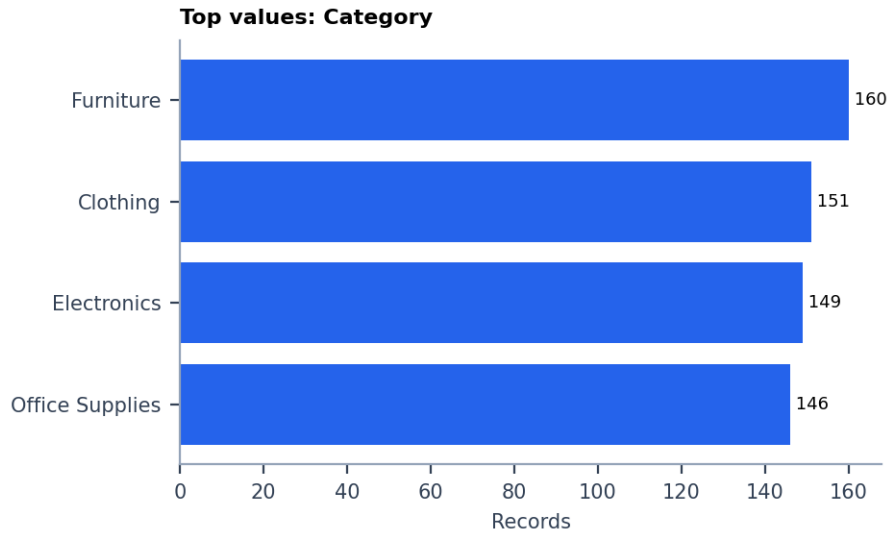
Distribution: Quantity



Data Completeness by Column



Category Composition



AI Data-Quality Narrative

38 cells are missing in total. The most affected column is 'Customer_Satisfaction' at 3.3% missing. Imputation and de-duplication in the Data Pipeline Studio are recommended before modelling.

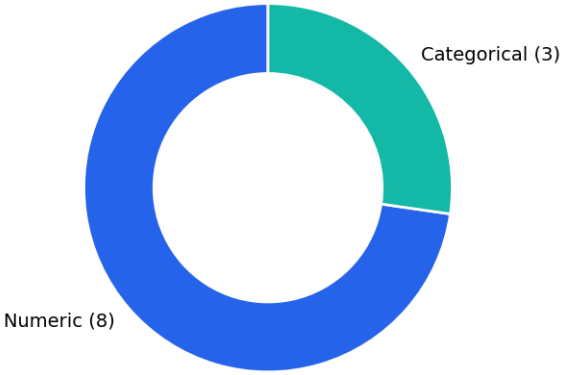
ETL actions available in the app: mean/mode or drop imputation, 5 scalers (Standard, MinMax, Robust, Quantile, Power), de-duplication, rare-category pooling.



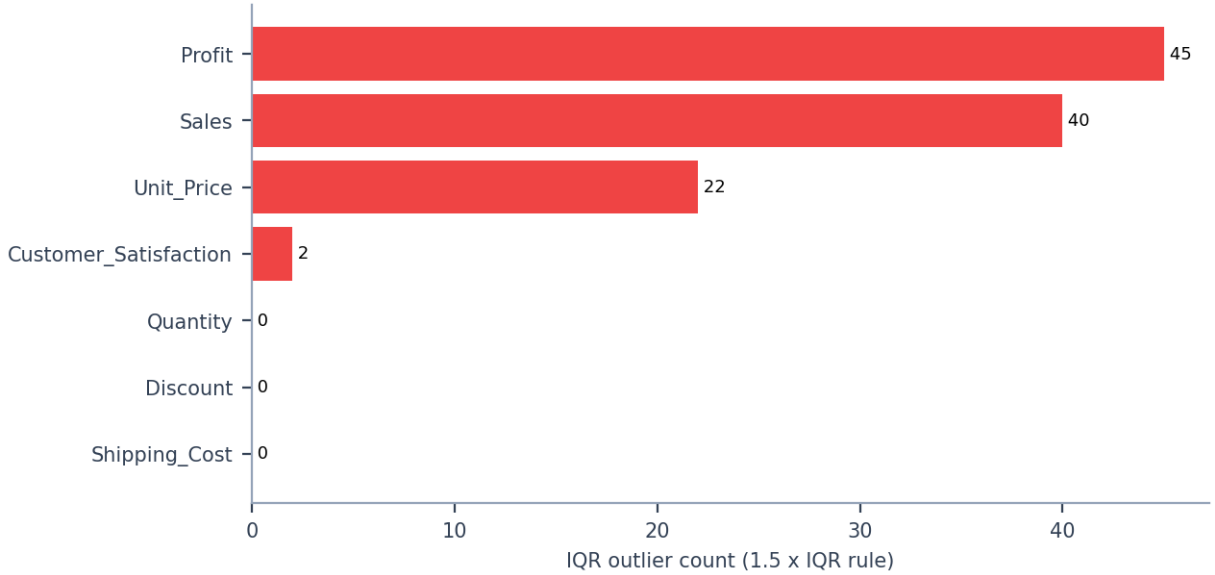
Missing Values (% by column)



Schema Composition



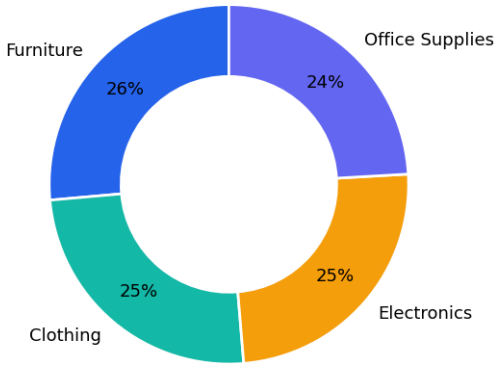
Outlier Scan (IQR rule) per numeric column



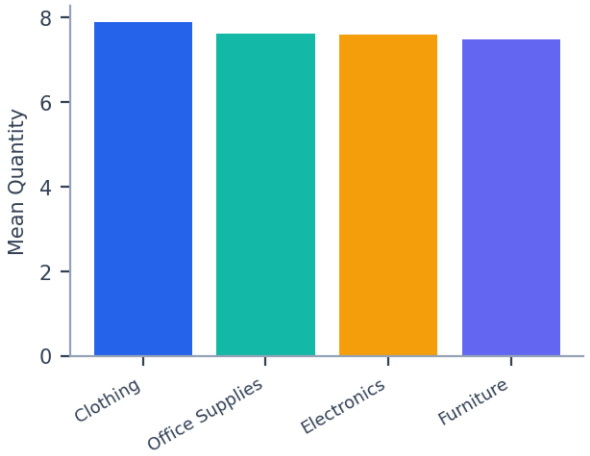
AI Narrative

The BI Canvas showcase combines a category share donut, comparative bars, a trend line, a scatter plot, a data-completeness gauge and a waterfall contribution chart so stakeholders can read composition, trend and relationship patterns at a glance.

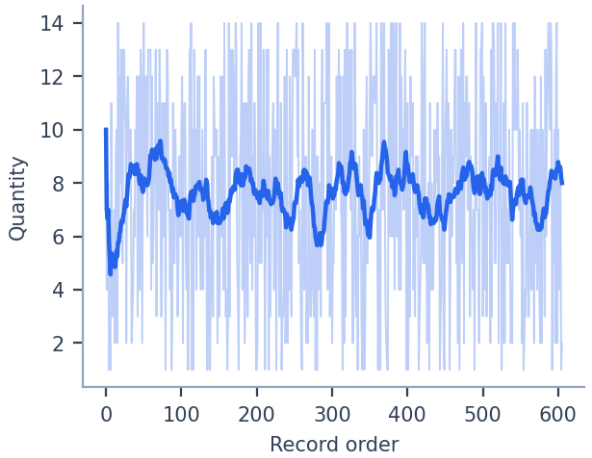
Donut Chart



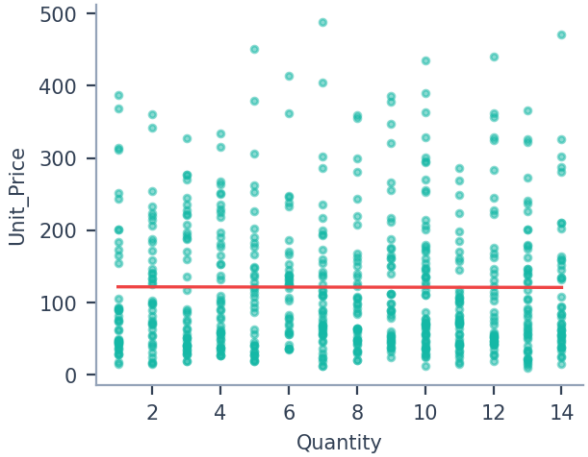
Clustered Bar Chart



Line Chart (trend)



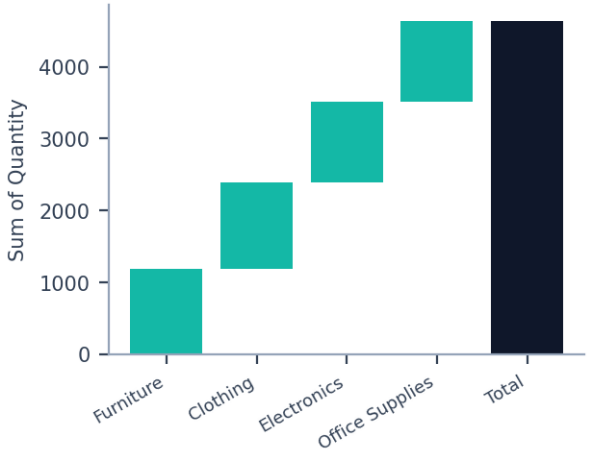
Scatter Plot



Gauge (data completeness)



Waterfall Chart



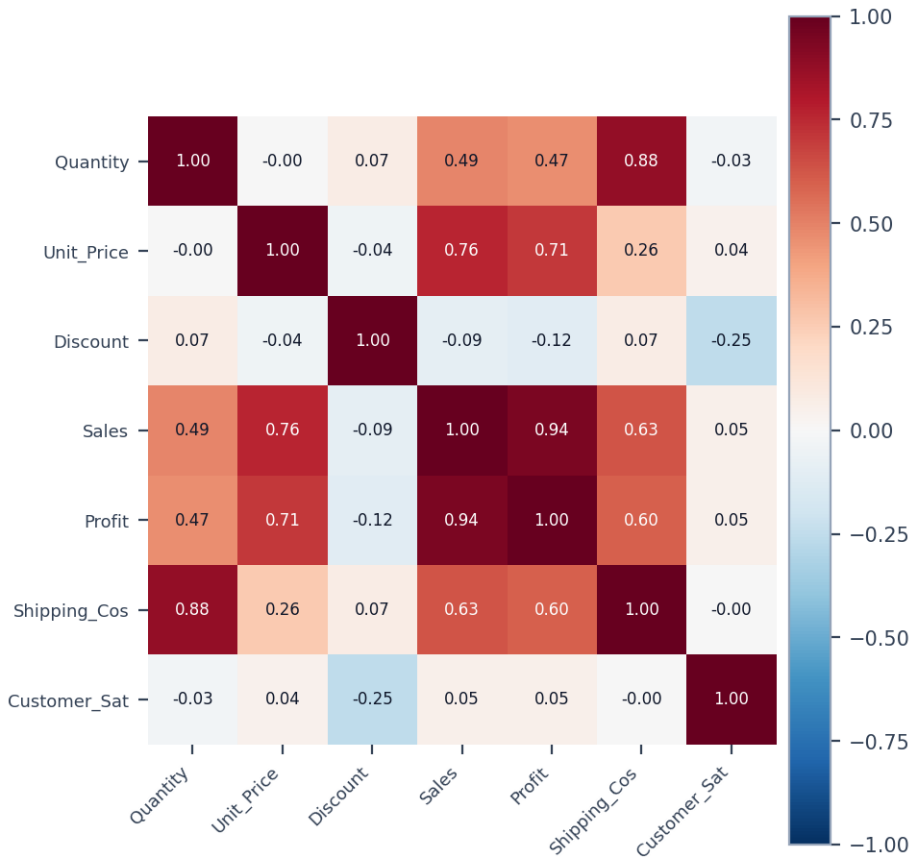
Relationships & Descriptive Statistics

Correlation matrix, strongest pairs and numeric profile

AI Narrative

The strongest linear relationship is between 'Sales' and 'Profit' (r = 0.941), a positive association. Highly correlated pairs may carry redundant information for modelling.

Correlation Heatmap (Pearson)



Strongest Correlations

Variable A	Variable B	r	Strength
Sales	Profit	+0.941	Strong
Quantity	Shipping_Cost	+0.875	Strong
Unit_Price	Sales	+0.762	Strong
Unit_Price	Profit	+0.707	Strong
Sales	Shipping_Cost	+0.633	Moderate

Numeric Profile

Column	Mean	Std	Skew	Outliers
Quantity	7.64	4.02	-0.07	0
Unit_Price	121.54	93.46	1.26	22
Discount	0.06	0.07	0.98	0
Sales	899.81	936.39	2.19	40
Profit	156.21	183.68	2.56	45
Shipping_Cost	26.41	10.14	-0.06	0
Customer_Satisfaction	3.68	0.80	-0.29	2

AI Narrative

Shapiro-Wilk was run on 4 numeric columns and 0 look approximately normal. 4 of 4 ANOVA tests and 0 of 3 chi-square tests were significant at the 5% level.

Normality (Shapiro-Wilk)

Column	W	p	Normal?
Quantity	0.944	0.0000	No
Unit_Price	0.876	0.0000	No
Discount	0.791	0.0000	No
Sales	0.779	0.0000	No

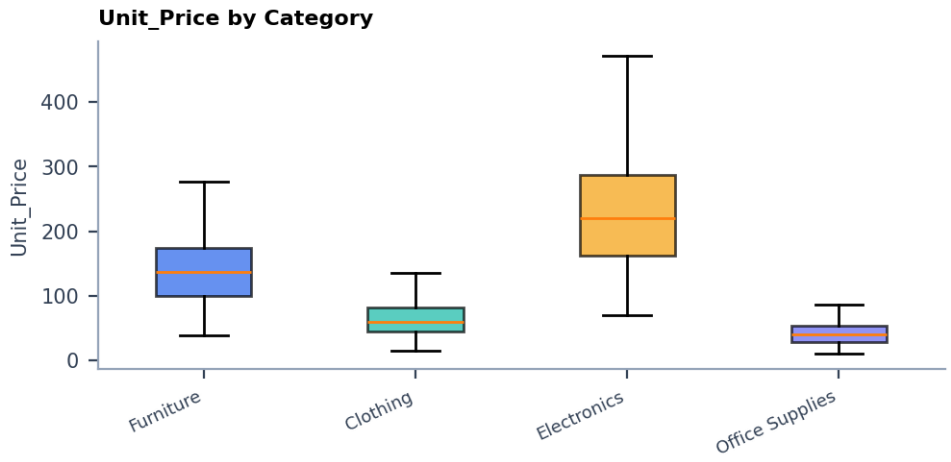
One-way ANOVA (numeric ~ category)

Numeric	Category	F	p	Sig.
Unit_Price	Category	333.66	0.0000	Yes
Sales	Category	104.98	0.0000	Yes
Profit	Category	86.50	0.0000	Yes
Sales	Region	3.27	0.0210	Yes

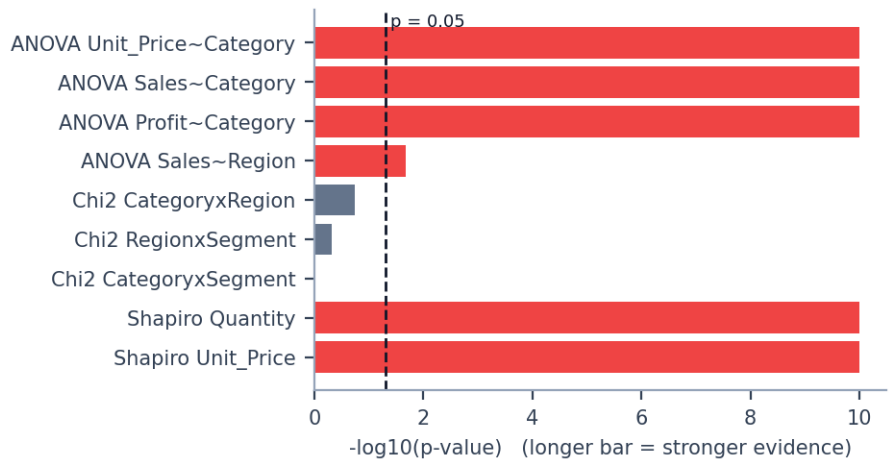
Chi-square Association

A	B	V	p
Category	Region	0.084	0.1820
Region	Segment	0.068	0.4805
Category	Segment	0.030	0.9823

Group Comparison (top ANOVA pair)



Evidence Strength Across All Tests



Predictive ML Lab

Auto-trained regression and classification baselines (80/20 hold-out)

AI Narrative

For regression on 'Sales', Linear Regression achieved R2 = 0.9308 and RMSE = 277.7664; the most influential feature is 'Profit'. Classification of 'Category' reached accuracy 0.5492 (weighted F1 0.5498).

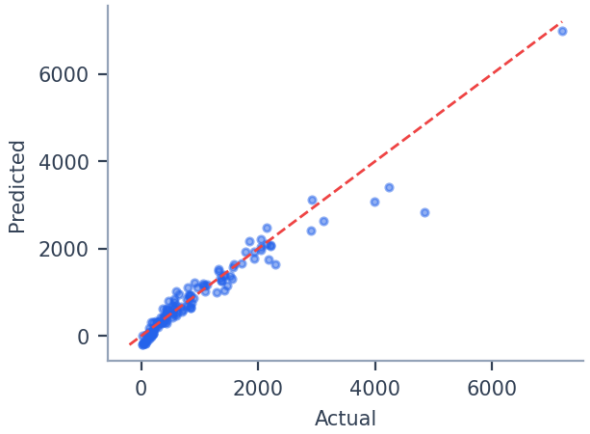
REGRESSION TARGET
Sales
Linear Regression

R2 / RMSE
0.931 / 277.77
hold-out set

CLASSIFICATION TARGET
Category
Random Forest

ACCURACY / F1
0.549 / 0.550
weighted F1

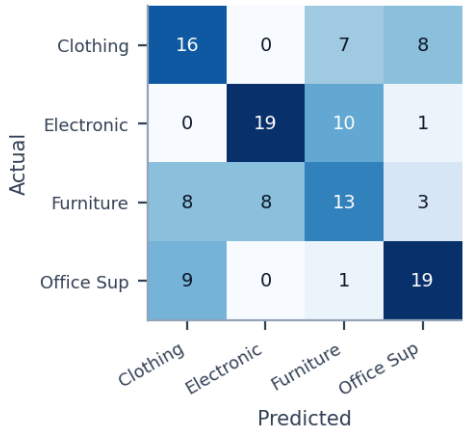
Predicted vs Actual



Feature Importance



Confusion Matrix



Regression Model Comparison

Model	R2	RMSE	MAE
Linear Regression	0.931	277.77	164.57
Ridge	0.931	277.94	164.10
Random Forest	0.906	324.36	165.14

Classification Model Comparison

Model	Accuracy	Weighted F1
Logistic Regression	0.549	0.549
Random Forest	0.549	0.550

Recommended Actions

1. Remove duplicate rows before reporting to avoid double counting.
2. Investigate the source of missing values in 'Customer_Satisfaction' and apply an imputation policy.
3. Review the 45 outliers in 'Profit' to separate errors from genuine extremes.
4. Focus business levers on 'Profit', the strongest driver of 'Sales'.
5. Re-run this report after each data refresh to track quality and model drift.

Key Findings (computed)

- The dataset contains 606 rows and 11 columns (8 numeric, 3 categorical).
- Data completeness is 99.43% with 38 missing cells and 6 duplicate rows.
- Most incomplete column: 'Customer_Satisfaction' (3.3% missing).
- Most volatile metric: 'Profit' (coefficient of variation 1.1759).
- 'Profit' has the most IQR outliers (45).
- Strongest relationship: 'Sales' vs 'Profit' (r = 0.941).
- 'Category' significantly affects 'Unit_Price' (ANOVA p = 0.0).
- Best regression model for 'Sales': Linear Regression (R2 = 0.9308, RMSE = 277.7664).
- Best classifier for 'Category': Random Forest (accuracy = 0.5492, weighted F1 = 0.5498).

About this Narrative

Generated by the offline rule engine
Provider: Offline (rule-based, no API key)
Model: rule-engine-v1
Privacy: only aggregated statistics (counts, means, test results) are sent to the LLM; raw rows never leave the app.

Platform Feature Map

Every capability of the suite and what it found in this dataset

Test Report

15 Oct 2026, 12:31

Capabilities

Feature	Where in the app	What it does	Result on this dataset
CSV ingestion & cross-filtering	Sidebar	Upload a CSV, filter by any category, slice any numeric range; all tabs update.	606 rows x 11 columns loaded
Operations Overview	Tab 1	KPI cards, live data monitor and algorithmic insight engine.	99.43% complete, 6 duplicates
Data Pipeline Studio (ETL)	Tab 2	Imputation, 5 scalers, de-duplication, noise pooling, schema profiling.	38 missing cells flagged for imputation
Native BI Canvas	Tab 3	15 visuals: bar, column, line, pie, donut, treemap, scatter, table, matrix, card, KPI, gauge, waterfall, funnel, map.	About 14 of 15 visuals applicable to this schema
Diagnostic Statistics	Tab 4	Shapiro-Wilk, ANOVA / Kruskal-Wallis, chi-square association tests.	11 tests run, 4 significant association(s)
Predictive ML Lab	Tab 5	Regression (OLS, Ridge, Lasso, RF) and classification (LogReg, RF, GB) with K-fold CV.	R2 0.931 on 'Sales'; F1 0.550 on 'Category'
Multimodal CV Auditor	Sidebar mode 2	Image upload with luminosity / dark-field diagnostics.	Image-based; not applicable to CSV data
LLM Narrator	Tab 6	Provider-agnostic narrator (Claude, OpenAI, Groq, Gemini, offline) writing every section's commentary.	Offline (rule-based, no API key) / rule-engine-v1
Dashboard PDF Publisher	Tab 6	Compiles all of the above into this 8-page landscape dashboard.	You are reading it

Methodology & Notes

Outliers use the 1.5 x IQR rule. Correlations are Pearson (ID-like columns excluded). Shapiro-Wilk uses up to 5,000 sampled values. ANOVA is run for categorical columns with 2-10 levels; chi-square for pairs with up to 15 levels (Cramer's V = effect size). ML baselines use an 80/20 random split (seed 42); the regression target is the numeric column most correlated with the others and the classification target is the first categorical column with 2-10 classes. These are automatic baselines for orientation; use the Predictive ML Lab tab for tuned, user-selected models.